

COORDINATE-BLOCK FACTORIZATION OVER STRONGLY PURELY INFINITE RINGS AND APPLICATIONS TO PROPERTY $(TT)/T$

GPT-5.6 SOL

ABSTRACT. We prove a coordinate-block factorization theorem for general linear groups over rings satisfying the single-sandwich condition

$$a \neq 0 \implies xay = 1 \text{ for some } x, y.$$

Every element of $\mathrm{GL}_n(R)$ is a product of at most $2n + 6$ factors, each an elementary transvection or an element of one fixed coordinate copy of $\mathrm{GL}_{n-1}(R)$. Thus $\mathrm{GL}_n(R)$ has width at most $2n + 6$ relative to the elementary root subgroups and that fixed coordinate block.

We combine this algebraic factorization with relative quasi-cocycle rigidity for elementary groups over finite free associative $\mathbb{F}_2$ -algebras. If R is a nontrivial finite-type $\mathbb{F}_2$ -algebra carrying a binary Leavitt family, satisfying the single-sandwich condition, and satisfying $\mathrm{diag}(u, 1) \in E_2(R)$ for every $u \in R^\times$, then $E_n(R)$ has property $(TT)/T$ for every $n \geq 3$. Consequently,

$$E_n(L_{\mathbb{F}_2}(1, 2)) \text{ has property } (TT)/T \quad (n \geq 3).$$

The proof has two complementary parts: a quantitative root quasi-cocycle estimate and a Gaussian-elimination argument based on the single-sandwich property. We also give a direct tree-theoretic factorization of the unstable diagonal matrices needed for the Leavitt specialization. Together with the previously established nonsocifity of these elementary Leavitt groups, the result gives explicit nonsocif groups with property $(TT)/T$.

1. INTRODUCTION

Property $(TT)/T$, introduced by Mimura, asks that Hilbert-space quasi-cocycles be bounded after the invariant-vector part of the representation has been removed. It is stronger than Kazhdan's property (T) and is designed to control bounded cohomology and homomorphisms into groups admitting unbounded quasi-cocycles [5]. Ershov and Jaikin-Zapirain proved property (T) for elementary groups over arbitrary finitely generated associative rings [3]. The corresponding quasi-cocycle problem is subtler: the bounded error in the

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cocycle identity accumulates under a long word, so finite generation alone gives no global bound.

The present paper supplies a globalization mechanism for a class of strongly purely infinite rings. Its algebraic input is an explicit matrix factorization. The hypothesis is the following strong two-sided division property:

$$(1) \quad a \neq 0 \implies \exists x, y \in R, \quad xay = 1.$$

It is often called the single-sandwich property. It is available in the purely infinite simple setting relevant here [2].

Theorem A (Coordinate-block factorization). *Let R be a nontrivial ring satisfying (1), let $n \geq 2$, and fix a coordinate $j < n$. Every $A \in \mathrm{GL}_n(R)$ admits a factorization of length at most $2n + 6$ in which every factor is either an elementary transvection or belongs to the coordinate copy*

$$\mathrm{GL}_{n-1}^{(j)}(R) = \{B \in \mathrm{GL}_n(R) : Be_j = e_j, e_j^t B = e_j^t\}.$$

Theorem A is a relative bounded-width statement rather than bounded elementary generation: the bounded factors come from the elementary root subgroups together with one fixed coordinate block. This is exactly the form required by the quasi-cocycle argument.

For the synthesis theorem, a *binary Leavitt family* in R means elements x_0, x_1, y_0, y_1 satisfying

$$y_i x_j = \delta_{ij}, \quad x_0 y_0 + x_1 y_1 = 1.$$

It supplies the matrix self-similarity used to move between ranks.

Theorem B (Rigidity synthesis). *Let R be a nontrivial finite-type $\mathbb{F}_2$ -algebra. Assume that*

- (1) R contains a binary Leavitt family;
- (2) R satisfies the single-sandwich property (1);
- (3) $\mathrm{diag}(u, 1) \in E_2(R)$ for every $u \in R^\times$.

Then $E_n(R)$ has property $(TT)/T$ for every $n \geq 3$.

The assumptions play separate roles: finite type presents the ring as a quotient of a finite free algebra, the Leavitt family moves between matrix ranks, the single-sandwich property supplies bounded matrix reduction, and the diagonal condition identifies elementary and general linear groups.

Theorem C (Binary Leavitt algebra). *For every $n \geq 3$,*

$$E_n(L_{\mathbb{F}_2}(1, 2)) \quad \text{has property } (TT)/T.$$

Combining Theorem C with the independently known nonsocicity theorem gives the following consequence.

Corollary 1.1. *For every $n \geq 3$, the group $E_n(L_{\mathbb{F}_2}(1, 2))$ is nonsocic and has property $(TT)/T$.*

Corollary 1.1 combines two logically independent results: Theorem C supplies property $(TT)/T$, while the previous nonsocicity theorem supplies the other assertion.

Relation to previous work. The argument combines several established theories. Mimura introduced property $(TT)/T$, developed the relative-to-global strategy for quasi-cocycles, and proved the relative results for universal-lattice settings used below [5, 6]. Ershov and Jaikin-Zapirain proved property (T) for elementary groups over finitely generated associative rings [3]. On the ring-theoretic side, Ara, Goodearl, and Pardo established the GE and unit-group K_1 structure of purely infinite simple rings [2], while Ara, Brustenga, and Cortiñas computed the stable algebraic K -theory of Leavitt path algebras [1].

Within this framework, the new algebraic input is the fixed-coordinate factorization of Theorem A, including its explicit width $2n + 6$. The rigidity theorem arises by using that factorization to globalize relative quasi-cocycle bounds. Sections 2 and 5 also give direct quantitative constructions for the root estimate and the unstable diagonal reduction, respectively; the corresponding qualitative facts can also be recovered from the earlier structural results just cited.

Organization. Section 2 develops the relative root estimate. Section 3 proves the five-move pivot and the coordinate-block factorization. Section 4 combines the analytic and algebraic inputs to prove Theorem B, and Section 5 verifies its hypotheses for the binary Leavitt algebra. The final section discusses the natural extensions suggested by the proof.

2. QUASI-COCYCLES AND RELATIVE CONTROL

Let G be a group and let $\pi : G \rightarrow \mathcal{U}(\mathcal{H})$ be a unitary representation on a complex Hilbert space. A map $b : G \rightarrow \mathcal{H}$ is a *quasi-cocycle of defect at most D* if

$$(2) \quad \|b(gh) - b(g) - \pi(g)b(h)\| \leq D \quad (g, h \in G).$$

An exact cocycle has $D = 0$; it is the translation part of an affine isometric action. A quasi-cocycle permits uniformly bounded failure of the affine action law.

Definition 2.1. A group G has property $(TT)/T$ if, for every unitary representation π with no nonzero invariant vector, every Hilbert-space quasi-cocycle into π is bounded on G .

Proposition 2.2 (Mimura’s relative input). *Let X be finite and $A = \mathbb{F}_2\langle X \rangle$. The pair*

$$(E_4(A), X_{03})$$

has relative property $(TT)/T$.

Proof. Mimura proves relative $(TT)/T$ for the noncommutative universal lattice pair in [6, Proposition 9.2.8]:

$$E_m(\mathbb{Z}\langle X \rangle) \geq E_{m-1}(\mathbb{Z}\langle X \rangle), \quad m \geq 3$$

Relative $(TT)/T$ descends through a surjective homomorphism: pull a representation and quasi-cocycle back along the quotient map and then apply the relative bound. Reduce coefficients modulo 2, take $m = 4$, and conjugate the standard rank-three block so that it contains X_{03} . Boundedness on that block implies boundedness on the root subgroup. $\square$

The local input used below is slightly stronger than a relative $(TT)/T$ statement because no hypothesis is imposed on invariant vectors.

Theorem 2.3 (Root quasi-cocycle estimate). *Let X be finite, put $A = \mathbb{F}_2\langle X \rangle$, and let*

$$G = E_4(A), \quad X_{03} = \{I + aE_{03} : a \in A\}.$$

For every unitary representation π , every $D \geq 0$, and every quasi-cocycle $b : G \rightarrow \mathcal{H}$ of defect at most D , the set $b(X_{03})$ is bounded.

Proof. Let N be the last-column subgroup generated by X_{03}, X_{13}, X_{23} . Decompose

$$\mathcal{H} = \mathcal{H}^N \oplus (\mathcal{H}^N)^\perp, \quad b(r) = p + w$$

for $r = I + aE_{03}$. Since the coefficient ring has characteristic two, $r^2 = 1$. Applying (2) to r^2 , and projecting to $\mathcal{H}^N$, bounds the fixed component:

$$(3) \quad \|p\| \leq D.$$

For the moving component, let Q_X be the finite rank-three control set used in the two coordinate embeddings and put

$$M = \sum_{s \in Q_X} (\|b(\iota_{123}(s))\| + \|b(\iota_{012}(s))\|), \quad B = 2M + 2D,$$

$$K = 2(6|X| + 6)(B + 1).$$

If a control element s commutes with r and normalizes N , comparison of the quasi-cocycle equations for sr and rs , followed by orthogonal projection, gives

$$\|\pi(s)w - w\| \leq \|\pi(s)b(r) - b(r)\| \leq 2\|b(s)\| + 2D \leq B.$$

A quantitative rank-three plane estimate propagates these finitely many inequalities to the complete root planes in the two embeddings. In particular, for every $c \in A$,

$$\|\pi(x_{13}(c))w - w\|, \quad \|\pi(x_{23}(c))w - w\|, \quad \|\pi(x_{01}(c))w - w\| \leq K.$$

The Steinberg relation

$$[I + cE_{01}, I + E_{13}] = I + cE_{03}$$

and the unitary commutator estimate give displacement at most $4K$ under $x_{03}(c)$. Every element of N is a product of one element from each of X_{03}, X_{13}, X_{23} , so its displacement of w is at most $6K$. Since $w \perp \mathcal{H}^N$, the uniform-orbit projection lemma gives $\|w\| \leq 6K$. Together with (3), this gives

$$\|b(I + aE_{03})\| \leq D + 12(6|X| + 6)(2M + 2D + 1),$$

uniformly in $a \in A$. The numerical constant is not optimized. $\square$

Remark 2.4 (A qualitative proof from Mimura's theorem). For qualitative boundedness, Proposition 2.2 gives a shorter proof of Theorem 2.3. Split $\mathcal{H} = \mathcal{H}^G \oplus (\mathcal{H}^G)^\perp$. Mimura's relative theorem bounds the second projection on X_{03} . On the invariant summand the action is trivial, while $r^2 = 1$ for $r \in X_{03}$; the quasi-cocycle identities for 1 and r^2 then give $\|b_0(r)\| \leq D$. The direct proof above complements this argument by recording an explicit bound and the quantitative rank-three propagation behind it.

3. COORDINATE-BLOCK FACTORIZATION

We now prove the main algebraic input. Write

$$e_{ij}(a) = I + aE_{ij} \quad (i \neq j)$$

for an elementary transvection.

Lemma 3.1 (Five-move pivot). *Let R satisfy (1), let $A \in \mathrm{GL}_n(R)$, and fix distinct coordinates i, j . At most two elementary row operations and at most three elementary column operations produce a matrix B satisfying*

$$B_{jj} = 1, \quad B_{ij} = B_{ji} = 0.$$

Proof. We give the calculation, since it is the source of the uniform constant. Choose two distinct indices i, j . First arrange that the entry $a = A_{ii}$ is nonzero. If it is zero, invertibility of A gives k with $A_{ik} \neq 0$; right multiplication by $e_{ki}(1)$ replaces A_{ii} by A_{ik} . This costs at most one preliminary column operation.

Write $b = A_{ij}$, and choose $p, q \in R$ with $paq = 1$. Put

$$\rho = q(1 - pb), \quad F_1 = e_{ij}(\rho), \quad A_1 = AF_1, \quad b_1 = (A_1)_{ij}.$$

Then

$$b_1 = a\rho + b, \quad pb_1 = paq(1 - pb) + pb = 1.$$

Set

$$d_1 = (A_1)_{jj}, \quad w = (1 - d_1)p, \quad E_1 = e_{ji}(w), \quad A_2 = E_1A_1.$$

The (j, j) -entry is now

$$(A_2)_{jj} = d_1 + wb_1 = d_1 + (1 - d_1)pb_1 = 1.$$

Next set

$$E_2 = e_{ij}(-b_1), \quad A_3 = E_2A_2.$$

This clears (i, j) without changing the new 1 at (j, j) . Finally, with $c_3 = (A_3)_{ji}$, put

$$F_2 = e_{ji}(-c_3), \quad A_4 = A_3F_2.$$

This clears (j, i) , again preserving $(j, j) = 1$ and $(i, j) = 0$. Thus the four main moves use two row and two column transvections; including the possible preliminary column addition gives the asserted bounds 2 and 3. No stable-range or commutativity hypothesis is used. $\square$

Theorem 3.2. *Under the hypotheses of Theorem A, every $A \in \text{GL}_n(R)$ can be written*

$$A = g_1 \cdots g_m, \quad m \leq 2n + 6,$$

where each g_i is an elementary transvection or lies in $\text{GL}_{n-1}^{(j)}(R)$.

Proof. Choose $i \neq j$. Lemma 3.1 gives elementary products E, F , represented by lists ℓ, r , such that

$$B = EAF, \quad B_{jj} = 1, \quad B_{ij} = B_{ji} = 0,$$

with

$$|\ell| \leq 2, \quad |r| \leq 3.$$

The last two displayed zeros are not needed for the subsequent clearing, but they are part of the exact pivot calculation.

For every $k \neq j$, right multiply by

$$e_{jk}(-B_{jk}).$$

These transvections have a common source column and distinct target columns, so they do not interfere with one another. If T is their product and $C = BT$, then

$$C_{jc} = \begin{cases} 1, & c = j, \\ 0, & c \neq j. \end{cases}$$

The list t of right-clearing factors has length at most n .

For every $k \neq j$, left multiply by

$$e_{kj}(-C_{kj}).$$

Again the operations have distinct target rows and do not interfere. If S is their product and $D = SC$, then

$$D_{rc} = \begin{cases} 1, & r = c = j, \\ 0, & r = j, c \neq j, \\ 0, & r \neq j, c = j, \\ C_{rc}, & r, c \neq j. \end{cases}$$

Hence $D \in \text{GL}_{n-1}^{(j)}(R)$. The list s of left-clearing factors also has length at most n .

Solving $D = SEAF$ for A gives

$$A = E^{-1}S^{-1}DT^{-1}F^{-1}.$$

At the word level, inverse products are represented by reversing the factor lists and inverting every transvection. The inverse of $e_{ab}(z)$ is $e_{ab}(-z)$, so all these factors remain elementary. The resulting word has the exact form

$$\ell_{\text{rev}}^{-1} s_{\text{rev}}^{-1} [D] t_{\text{rev}}^{-1} r_{\text{rev}}^{-1},$$

and therefore length at most

$$2 + n + 1 + n + 3 = 2n + 6.$$

□

Remark 3.3. The fixed coordinate j is chosen before A . Allowing the coordinate block to depend on A would give a weaker statement and would not be sufficient for the uniform quasi-cocycle globalization used below.

4. FROM RELATIVE CONTROL TO GLOBAL RIGIDITY

Let R be a finite-type $\mathbb{F}_2$ -algebra. Choose a finite set X and a surjection

$$\mathbb{F}_2\langle X \rangle \twoheadrightarrow R.$$

Elementary matrices are functorial in the coefficient ring, so the root estimate of Theorem 2.3 descends to R . Coordinate permutations then give the same estimate for every root subgroup in rank four.

Two larger pieces are used in the assembly: a last-column subgroup and a last-row subgroup. The quasi-cocycle is bounded on both. The stabilized rank-three coordinate group normalizes these pieces. Kazhdan control for the rank-three group converts boundedness on the normalized root system into boundedness on the coordinate group, provided the ambient representation has no invariant vectors. This is the only place where the quotient by the trivial representation enters.

Lemma 4.1 (Elementary equals general linear). *Assume that R has a binary Leavitt family, satisfies (1), and satisfies $\text{diag}(u, 1) \in E_2(R)$ for every $u \in R^\times$. Then*

$$E_n(R) = \text{GL}_n(R) \quad (n \geq 2).$$

Proof. The single-sandwich elimination reduces an arbitrary element of $\text{GL}_2(R)$, by elementary row and column operations, to $\text{diag}(u, 1)$ for some $u \in R^\times$. The diagonal hypothesis therefore gives $\text{GL}_2(R) = E_2(R)$. A binary Leavitt family gives prefix-code coefficient isomorphisms $M_m(R) \cong R$. Apply the rank-two equality after transporting coefficients to $M_n(R)$, flatten the resulting block matrices, swap the two finite index factors, unflatten in the other direction, and transport coefficients back through $M_2(R) \cong R$. This is the coefficient/block argument which gives the assertion in every rank $n \geq 2$. □

Proposition 4.2 (Rank-four assembly). *Under the hypotheses of Theorem B, $E_4(R)$ has property $(TT)/T$.*

Proof. Choose a finite free-algebra presentation of R . Theorem 2.3 descends through this quotient and gives relative $(TT)/T$ for $X_{03}(R)$. Coordinate permutations give it for the six roots

$$X_{03}, X_{13}, X_{23}, X_{30}, X_{31}, X_{32}.$$

Property (T) for $E_3(R)$ follows from Ershov–Jaikin–Zapirain, or equivalently by descending the finite Kazhdan system for the free algebra. The Leavitt rank equivalence transports this to $E_4(R)$. The standard rank-four parabolic argument now combines the six relative root bounds: the last-row

and last-column pieces are bounded, their normalized rank-three core has almost-invariant quasi-cocycle values, and the Kazhdan estimate bounds those values. Hence $E_4(R)$ has relative $(TT)/T$ with respect to the stabilized $E_3(R)$.

By Lemma 4.1, the elementary groups in ranks three and four are the full general linear groups. Under these identifications the stabilized core is exactly the coordinate block $GL_3^{(3)}(R)$. Finally, coordinate permutations carry this relative statement to $GL_3^{(j)}(R)$ for every $j \in \{0, 1, 2, 3\}$.

Each elementary transvection lies in a coordinate block avoiding its two indices. Hence the union of one fixed coordinate block and all root subgroups is bounded for the quasi-cocycle. Theorem A writes every element of $GL_4(R)$ as a product of at most (14) elements of that union. Lemma 4.3 below, or directly the corresponding bounded-product lemma, makes the quasi-cocycle bounded on $GL_4(R)$. Transporting through $E_4(R) = GL_4(R)$ proves the proposition. $\square$

The passage from a bounded generating subset to the whole group uses the following elementary estimate.

Lemma 4.3 (Products of controlled pieces). *Let b be a quasi-cocycle of defect at most D , and suppose $\|b(g_i)\| \leq C$ for a list $(g_1, \dots, g_m)$. Then*

$$\|b(g_1 \cdots g_m)\| \leq mC + (m+1)D.$$

Proof. Induct on the list. The empty-list case is $\|b(1)\| \leq D$. For a nonempty list, apply (2) to the head and the product of the tail, use that the representation is isometric, and invoke the inductive bound. The displayed estimate is a slightly loose uniform form that also covers the empty list. $\square$

Proof of Theorem B. Proposition 4.2 gives the result in rank four. For arbitrary $n \geq 3$, the binary Leavitt family supplies an explicit group equivalence

$$E_n(R) \cong E_4(R).$$

Property $(TT)/T$ is invariant under group equivalence, so the rank-four conclusion transports to rank n . $\square$

5. THE BINARY LEAVITT SPECIALIZATION

Let L be the binary Leavitt algebra [4],

$$L = L_{\mathbb{F}_2}(1, 2) = \mathbb{F}_2\langle x_0, x_1, y_0, y_1 \rangle / (y_i x_j - \delta_{ij}, x_0 y_0 + x_1 y_1 - 1).$$

The defining generators themselves form a binary Leavitt family, and the presentation makes L a finite-type $\mathbb{F}_2$ -algebra. Pure infiniteness and simplicity give the single-sandwich property [2]. It remains to discuss the unstable diagonal condition.

Proposition 5.1 (Unstable diagonal reduction). *For every $u \in L^\times$,*

$$\text{diag}(u, 1) \in E_2(L).$$

Consequently $GL_2(L) = E_2(L)$.

The proposition has two complementary proofs. The first places it inside the known structure theory of purely infinite simple rings; the second produces the elementary factors directly.

5.1. A proof from structural results. Ara–Brustenga–Cortiñas give, for a regular coefficient ring, the exact sequence determined by $1 - N_E^t$ [1]. For the graph with one vertex and two loops this map is multiplication by -1 , hence

$$K_i(L_k(1, 2)) = 0$$

in every degree covered by their theorem, in particular in degree one.

There is also an unstable route already implicit in the structural literature. Ara–Goodearl–Pardo prove that a unital purely infinite simple ring is a GE-ring and that

$$K_1(R) \cong R_{\text{ab}}^\times$$

This is Theorem 2.3 of [2]. Thus $K_1(L) = 0$ makes each unit a product of commutators. The standard 2×2 Whitehead identity places $\text{diag}([a, b], 1)$ in $E_2(L)$; the GE property then gives $\text{GL}_2(L) = E_2(L)$, proving Proposition 5.1.

5.2. A direct tree-and-pencil factorization. There is also a direct tree-and-pencil reduction. Finite prefix codes encode the self-similar pieces of a unit. Elementary row and column moves refine the two codes to a common tree, isolate code-scalar blocks, and reduce the remaining one-sided factor. The output is an actual finite list of elementary matrices whose product is $\text{diag}(u, 1)$.

Proposition 5.2 (Finite factor witness). *For every $u \in L^\times$, there exists a finite list $(e_1, \dots, e_m)$ of elementary 2×2 matrices such that*

$$\text{diag}(u, 1) = e_1 \cdots e_m.$$

Thus the direct proof is effective for each given unit. Extracting uniform word-length or complexity estimates from the construction is a separate quantitative problem.

Proof of Theorem C. The binary Leavitt algebra is a nontrivial finite-type $\mathbb{F}_2$ -algebra, its canonical generators form a binary Leavitt family, pure infiniteness supplies (1), and Proposition 5.1 supplies the elementary diagonal condition. Theorem B applies. $\square$

6. FURTHER DIRECTIONS

The characteristic-two hypothesis enters most visibly in (3), where every root element has order two. Over a field of characteristic p , the corresponding root has order p , and iterating the quasi-cocycle identity gives

$$b(1) \approx b(r) + \pi(r)b(r) + \cdots + \pi(r)^{p-1}b(r).$$

On the root-fixed space this is p times the fixed projection of $b(r)$, with error at most $(p-1)D$. Thus the fixed-space argument has a plausible

finite-characteristic analogue. Extending the theorem requires uniform replacements for the characteristic-two rank-three control and the sign-free coordinate symmetries.

The algebraic factorization theorem suggests a separate question. For which natural classes of simple or purely infinite rings does the single-sandwich condition hold, and when can the coordinate block in Theorem A itself be controlled by elementary subgroups? A satisfactory answer would turn the present synthesis theorem into a broad rigidity criterion rather than a theorem whose most memorable instance is a Leavitt algebra.

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An accompanying Lean formalization is available [online](#).

REFERENCES

- [1] P. Ara, M. Brustenga, and G. Cortiñas, *K-theory of Leavitt path algebras*, Münster J. Math. **2** (2009), 5–34.
- [2] P. Ara, K. R. Goodearl, and E. Pardo, *K_0 of purely infinite simple regular rings*, K-Theory **26** (2002), no. 1, 69–100.
- [3] M. Ershov and A. Jaikin-Zapirain, *Property (T) for noncommutative universal lattices*, Invent. Math. **179** (2010), no. 2, 303–347.
- [4] W. G. Leavitt, *The module type of a ring*, Trans. Amer. Math. Soc. **103** (1962), 113–130.
- [5] M. Mimura, *Property (TT) modulo T and homomorphism superrigidity into mapping class groups*, Monatsh. Math. **178** (2015), no. 4, 573–600.
- [6] M. Mimura, *Rigidity theorems for universal and symplectic universal lattices*, doctoral thesis, University of Tokyo, 2011, [doi:10.15083/00004046](https://doi.org/10.15083/00004046).